

VIJAY NARAYAN REDDY MADIREDDY

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Visa Status: Permanent Resident of the United States

SUMMARY

Software Developer with experience in Cloud and Full-Stack development, proficient in Java, JavaScript and skilled in data structures and algorithms. Eager to leverage problem-solving skills and passion for efficient coding to contribute to innovative software solutions. My relevant coursework are Data Structures and Algorithms, Advanced Software Engineering, Object-Oriented Programming, Graph theory, Artificial Intelligence, Machine Learning, Natural Language Processing, Network Security.

EDUCATION

Michigan State University, East Lansing, MI

Aug 2024 - Dec 2025

Masters, Computer Science

•GPA: 3.928/4

Vellore Institute of Technology, Amaravati

Aug 2020 - Jul 2024

Bachelor's of Technology (B.Tech), Computer Science and Engineering

•GPA: 9.3/10

TECHNICAL SKILLS

- **Languages:** Java, Python, PHP, JavaScript, HTML, CSS, TypeScript, C# (.NET), Go
- **Libraries & Frameworks:** React.js, express.js, Flask, Bootstrap, node.js, THREE.js, Vite
- **Tools and Platforms:** Linux, Unity, Amazon Web Services (AWS) - EC2, Lex, Git, Docker, Postman, Figma, Jira, Azure - Containers, Resource Groups, Document Intelligence, Google Cloud Platform (GCP)
- **Databases:** MongoDB, SQL (Oracle, SQL Server), Redis, Firebase
- **Security & Pentesting:** Frida, Jadx, BurpSuite, Wireshark

EXPERIENCE

RiceFW Technologies

Jun 2025 - Sep 2025

Full-Stack Developer

Okemos, MI

- Automated document processing for Michigan State Police using Google Cloud, enhancing efficiency and accuracy.
- Contributed to Scan-AI software, automating form field extraction, resulting in a 40% reduction in manual data entry and a 25% increase in processing speed.
- Developed solutions using GCP Services such as Document AI, Storage, Workflows, VM, Cloud Run, and API, improving project scalability and integration.
- Utilized GCP **Python** APIs for prototyping workflows, streamlining development processes and enhancing project efficiency.

Zenoti

Jan 2024 - Jul 2024

Software Engineer Intern

- Collaborated within an **Agile**, cross-functional team of 40+ engineers on the Payments microservices platform.
- Developed, debugged, and deployed solutions using **ASP.NET (C#)**, **TypeScript**, and **JavaScript**, resolving 20+ critical bugs.
- Enhanced system stability by resolving over 20 critical bugs in the Point-of-Sale (POS) page, leading to a 15% improvement in system uptime.
- Engineered and deployed new features, including tax deposits with CheckHQ API and Adyen Terminal APIs, enhancing transaction processing efficiency.
- Acquired hands-on experience with microservices architecture, RESTful APIs, and CI/CD pipelines and IIS, which improved my ability to contribute to complex projects.

PROJECTS

Voice Typing - Browser Extension | *JavaScript*

- Built a browser extension that enables voice-to-text typing in any input field on the browser.
- Integrated Web Speech API with support for 10 languages (English, Telugu, Hindi, Spanish, German, Chinese, French, Japanese, Russian).
- Developed using JavaScript and Manifest V3, ensuring compatibility with modern Chrome extension standards.

ImmersiFair - Virtual Career Fair Platform | *THREE.js, React.ts, WebRTC, Bootstrap, express.js, websocket*

- A THREE.js interactive platform so that users can chat, talk (audio / video) and engage in the career fair event.
- Implemented real-time user position synchronization and chat using WebSockets with an Express.js backend.
- Created 3D assets using Blender for immersive experience.

LEGO Brick Object Detection | *Machine Learning, Python*

- Implemented object detection models such as YOLO, Fast R-CNN, Faster R-CNN, RetinaNet with ResNet-50 and ResNet-101 to detect LEGO bricks in the image with accuracies mAP greater than 94%.
- Implemented on NVIDIA H200 & A100 GPUs.

AI Scrutiny | *React.js, express.js, GSAP, Bootstrap, MongoDB, NGINX, Python, Flask*

- Led development of an AI-powered platform to automate answer sheet evaluation, reducing manual grading effort.
- Integrated Google Cloud Vision API for OCR to extract text from scanned sheets with high accuracy.
- Applied NLP techniques and cosine similarity to evaluate responses against professor-provided answers.
- Deployed and optimized the platform using NGINX and MongoDB, ensuring scalability and reliability.

Chat-GPT Whatsapp bot | *pywebio, Glitch webhook, OpenAI API*

- Using JavaScript and webhooks, I built a Glitch application that automatically responds to messages through the OpenAI API, retrieving and sending back results to the user.
- You can access chatgpt using your whatsapp.

ACHIEVEMENTS

- Outstanding Capstone Project: AI Scrutiny
- LeetCode Solved 450+ Problems & practiced Data Structures, Algorithms, Optimization, Dynamic programming, Graphs.
- HackerRank Certifications: Problem Solving, SQL, Java
- Google Qualified & Participated in Code Jam, Kick Start Round A, F, G 2022
- Meta Competed in Meta HackerCup 2022

CLUBS & CHAPTERS

Women in Open Source (WiOS) club

Sep 2022 - Jan 2023

Technical Lead

- Led a collaborative team, mentoring peers and junior members and redesigning the club website using Bootstrap.
- Organized and managed the technical fest and events at our University, driving collaboration across teams and Experienced with project management tools such as Jira.

EXTRA CURRICULAR ACTIVITIES

Pencil Art, Playing Billiards and Foosball

Vijay Narayan Reddy Madireddy

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October 24, 2025

Hiring Committee

College of Engineering – Robotics Department

University of Michigan

Ann Arbor, MI

Respected Hiring Committee,

I am writing to express my strong interest in the **Applications Programmer Assoc.** position within the College of Engineering's Robotics Department at the University of Michigan. As a **Master's student in Computer Science at Michigan State University** with hands-on experience in **software development, cloud technologies**, I am eager to contribute my technical expertise and collaborative mindset to the Michigan Engineering's mission is to provide scientific and technological leadership to the people of the world.

In my most recent role as a **Full-Stack Developer at RiceFW Technologies**, I developed automated document processing solutions for the **Michigan State Police**, leveraging **Google Cloud Platform's Document AI, Cloud Run, and Workflows**. This project reduced manual input time by 40%, while improving the scalability of the platform. This role strengthened by **requirement gathering skills, Cloud technologies and insights into Compliance through Vanta – which perfectly aligns with the role.**

Previously, as a **Software Engineer Intern at Zenoti**, I collaborated in an Agile environment with a 40+ member engineering team to maintain and enhance a payment microservices platform. I contributed to both frontend and backend systems using **C#, TypeScript, and JavaScript and SQL** resolving over 20 critical bugs and improving system uptime by 15%. The role strengthened my **debugging, testing, and communication abilities** - key for providing robust application support within the Robotics Department.

Beyond professional experience, I have developed projects that reflect my interest towards learning and prototyping new things. My work on **ImmersiFair**, a real-time 3D career fair platform built using **Three.js, React, and WebRTC**, showcases my ability to merge web technologies with interactive visualization and networking components. Similarly, my project **AI Scrutiny** demonstrates my ability to integrate **Flask, MongoDB, and NLP techniques** to develop NLP-based evaluation systems.

What excites me most about this opportunity is the chance to build tools that directly support research, education, and innovation at Michigan Engineering—a community dedicated to making technology more accessible and impactful. I am confident that my background in **software development, problem-solving, and system optimization**, combined with my strong motivation to learn, will allow me to contribute effectively to your team's goals.

Thank you for your time. I would be honored to bring my technical skills and enthusiasm to support the Robotics Department's software initiatives. I look forward to hearing from you and discussing how I can contribute to Michigan Engineering.

Sincerely,
Vijay Narayan Reddy Madireddy